LIFE

Charity begins at *Homo sapiens*

By Mark Buchanan

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IN THE aftermath of the Indian Ocean tsunami last year, people from the world's richest countries were falling over each other to make donations to help rebuild the lives of the survivors. Perhaps it was the conjunction of this terrible natural disaster with the consumerist orgy of Christmas that spurred so many of us to greater generosity. Whatever the reason, conspicuous donation suddenly became the vogue. Individuals, and even entire countries, competed to see who could send most money to people on the other side of the world whose identity they did not know and who they were highly unlikely ever to meet. What an odd species we are.

Not that *Homo sapiens* is the only species in which individuals bestow kindness on others. Many mammals, birds, insects and even bacteria do likewise. But their largesse tends to be reserved for their genetic relatives; this makes sense in evolutionary terms, because by helping someone who shares many of your genes you improve the chances of propelling this common DNA into the future. Humans are different, for we cooperate with complete genetic strangers – workmates, neighbours, anonymous people in far-off countries. Why on earth do we do that?

For several decades, researchers have had a possible explanation: apparently selfless acts are nothing of the kind, but are instead a clever way of promoting individual self-interest. When rivals meet again and again, for example, the rewards of cooperation can outweigh the costs of conflict, so getting along pays dividends. Scientists have also come to realise what philanthropists such as Getty and Gates have long known: that altruism does wonders for your reputation. But does cooperation always have self-interested roots? Some researchers are starting to have their doubts.

Over the past decade, experiments devised by Ernst Fehr of the University of Zurich in Switzerland, among others, have shown that many people will cooperate with others even when it is absolutely clear they have nothing to gain. A capacity

for true altruism seems to be a part of human nature. It is a heartening discovery, yet one that has also touched off a firestorm of debate.

The experiments at the centre of the controversy are as simple as they are illuminating. They ignore theory-based preconceptions about how individuals ought to behave and focus instead on finding out what they actually do when playing games in which there is real money at stake.

One of the most basic of these games is the "ultimatum game". An experimenter gives one of two players some cash, say $20, and asks that person, called the "proposer", to offer a fraction of it to the second player, called the "receiver", whose identity is hidden from the other player. The proposer can offer any amount they choose, from nothing up to the entire $20. The receiver then has the choice of accepting or rejecting the offer. If he or she accepts, the cash is shared according to the original offer. A rejection means that no one gets anything. The game is played just once.

For the receiver, self-interest would seem to dictate accepting the offer no matter how small it is, since getting something is better than getting nothing. Knowing this, a similarly self-interested proposer should offer as little as possible. But over the past decade or so, research on student volunteers has shown that proposers in such experiments typically offer anything from 25 to 50 per cent, while receivers tend to reject offers of less than 25 per cent.

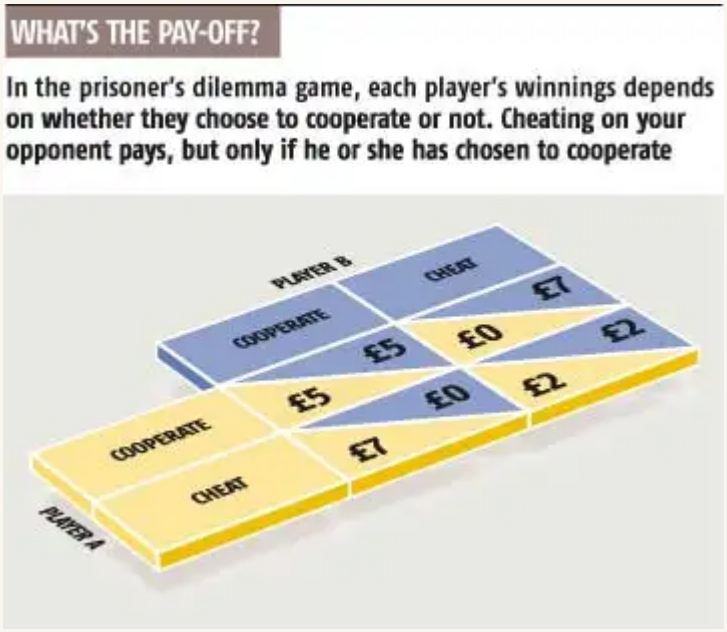
"People reject low offers," says anthropologist Joseph Henrich of Emory University in Atlanta, Georgia, "because they view them as unfair." And through their rejection, they show a willingness to punish the unfair offers even at a cost to themselves.

A vast number of other experiments illustrate the same point. Last year, for example, Fehr and his colleagues had students play a version of the famous prisoner's dilemma game, in which two people can prosper through cooperation but are also given strong incentives to cheat on one another. In this game, if the

participants cooperate, each receives a worthwhile monetary pay-off. But either player can get an even higher pay-off by cheating while their opponent cooperates (see Diagram).

WHAT'S THE PAY-OFF?

In the prisoner's dilemma game, each player's winnings depends on whether they choose to cooperate or not. Cheating on your opponent pays, but only if he or she has chosen to cooperate.



In this particular version of the game, the researchers got people to play sequentially: one would go and then the other, fully aware of what the first had done. In theory, anyone thinking only of their own personal gain would always cheat, as this pays more than cooperating. But in the experiments, although many of players who went first did cheat, others cooperated, despite knowing that the second player could sucker them by cheating. What's more, roughly half those who went second rewarded cooperation by treating their opponent fairly, even though that meant forgoing an easy pay-off for themselves (*Human Nature*, vol 13, p 1). "The facts are clear," Fehr says. "Many people are willing to cooperate and to punish those who don't, even when no gain is possible."

> *"Many people will cooperate, and punish those who don't, even when no gain is possible"*

This tendency – which researchers call **"strong reciprocity"** – throws into question the assumption that apparently selfless behaviour must have some selfish explanation. Across disciplines, researchers now agree that people often act against their own self-interest. "This is the most important empirical work on the human sense of justice in many years," says evolutionary biologist Robert Trivers of Rutgers University in New Jersey.

But when it comes to explaining the origin of our altruism, matters get a whole lot more contentious. In evolutionary terms it is a puzzle because any organism that helps others at its own expense stands at an evolutionary disadvantage. So if many people really are true altruists, as it seems, why haven't greedier, self-seeking competitors wiped them out?

One possibility, Trivers suggests, is that evolution actually is wiping these people out – it just hasn't finished the job yet. He, along with many anthropologists, takes the view that humans evolved to cooperate when our ancestors lived in small, isolated groups of hunter-gatherers. In this setting, they learned through repeated interaction with others that cooperation generally pays because it induces other members of the group to return a favour in the future. Biologists refer to strategic cooperation of this kind as "reciprocal altruism". It cannot directly explain the true altruism found in experiments in which anonymous players meet only once, offering them no hope of future gain. But it is the benefits we gained from reciprocal altruism in our evolutionary past that lead us to behave with "inappropriate" altruism in experiments like Fehr's, Trivers says. "Our brains misfire when presented with a situation to which we have not evolved a response."

If Trivers is right, then true altruism is what evolutionary biologists call a "maladaptation". Evolved to respond in a certain way to a given situation, we find it hard to act differently in the changed circumstances of the modern world. That would make strong reciprocity just another in a long list of maladaptations found in modern human behaviour, according to anthropologist John Tooby of the University of California at Santa Barbara. To make his point he gives the example of sexual desire, which most biologists agree evolved to spur the conception of

offspring. Today, however, individuals experience sexual desire in many situations in which procreation is clearly impossible, "even when they know the object of their desire is imaginary, or a piece of paper", as Tooby says.

> *"Any organism that helps others at its own expense stands at an evolutionary disadvantage"*

Undoubtedly adaptations that evolved to help us cope under specific conditions can backfire when situations change. But not everyone is convinced by the idea that true altruism is such a maladaptation. Henrich disagrees with the theory's central premise. He believes that while our ancestors lived in small, close-knit groups, one-shot interactions with strangers would have been common even then. What's more, these interactions could have been crucial to people's survival, because they would have occurred over shared resources such as water holes and prey animals and, more crucially, in times of catastrophe such as flood or drought. "Environmental shocks would have guaranteed that strangers encountered one another during fitness-critical times," Henrich says.

If both one-shot and repeated interactions were routine in ancestral life, Henrich argues, evolution would presumably have prepared us to distinguish between the two with some precision. And that does seem to be the case. Two years ago economists Simon Gächter of the University of St Gallen in Switzerland and Armin Falk of the University of Bonn, Germany, looked at how people alter the way they play the prisoner's dilemma game depending on whether the game involves one-shot or repeated encounters with others. If people treat one-shot encounters as if they were repeated – as the maladaptation idea suggests – then there shouldn't be a difference. But they found that repeated play more than doubled cooperation levels, indicating that we are fully capable of adapting our behaviour to the situation at hand (*The Scandinavian Journal of Economics*, vol 104, p 1).

Further support for the idea that strong reciprocity is an adaptation in its own right comes from the theoretical studies of economist Herbert Gintis of the University of Massachusetts, Amherst, anthropologist Robert Boyd of the

University of California at Los Angeles, and others. They set up a computer model in which groups of individuals interacted, and watched how their behaviour evolved. Individuals were set up in the model to behave initially either as cheats or as cooperators, and in personal interactions the former came off best. When groups competed with one another, however, cooperation came into its own: groups with more cooperators were likely to flourish.

But that was only the start. The individuals, whether initially cooperators or cheats, were also programmed to copy successful behaviour. In simulations with groups ranging from 4 to 256 individuals, the team found that altruism could evolve. The benefits that cooperation conferred on a group outweighed its costs to individuals – but only in groups of less than about 10. Ancestral human hunter-gatherer bands are thought to have numbered 30 or more individuals, so how could cooperative behaviour have evolved and spread in these groups?

The answer lies in the fact that strong reciprocity is not simply a matter of cooperation; it also requires punishment of those who fail to toe the line. When the team added punishment to their models, they found it made a huge difference. In a second round of simulations, they included a new kind of individual: the "punishers". These punishers were not only willing to cooperate with others but also to punish cheats. By making cheats pay for their antisocial actions, they tipped the balance towards cooperation. This time, competition between groups led to the emergence of cooperation in groups of up to 50 individuals (*Proceedings of the National Academy of Sciences*, vol 100, p 3531).

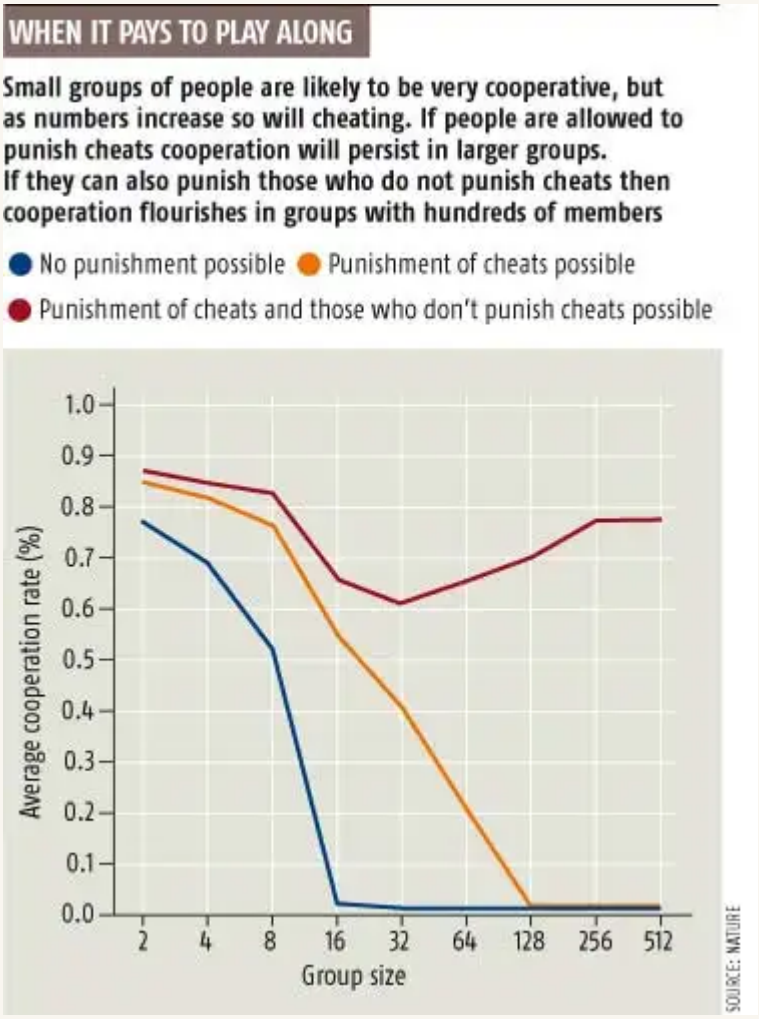
Could competition between small groups of our ancestors somehow have turned them into strong reciprocators? Gintis, Boyd and their colleagues believe so. What's more, subsequent research by Fehr, working with economist Urs Fischbacher of the University of Zurich, suggests that as humans came to live in larger groups, their attitudes towards reciprocity may have become even more hard-line. Using a similar model to Gintis and the others (*Nature*, vol 425, p 785), they found that cooperation can become the default behaviour in large groups provided punishers are willing to punish not only those who cheat, but also those

who fail to punish cheats (see Graph). "In this case," Fehr says, "even groups of several hundred individuals can establish cooperation rates of between 70 and 80 per cent."

WHEN IT PAYS TO PLAY ALONG

Small groups of people are likely to be very cooperative, but as numbers increase so will cheating. If people are allowed to punish cheats cooperation will persist in larger groups. If they can also punish those who do not punish cheats then cooperation flourishes in groups with hundreds of members.

- No punishment possible
- Punishment of cheats possible
- Punishment of cheats and those who don't punish cheats possible



These findings suggest that true altruism, far from being a maladaptation, may be the key to our species' success by providing the social glue that allowed our ancestors to form strong, resilient groups. It is still crucial for social cohesion in

today's very different world. "Something like it had to evolve," Gintis says.

In the absence of further discoveries, it seems likely that the argument over adaptation and maladaptation will continue. But this controversy is not the most important issue, says anthropologist Laurent Keller of the University in Lausanne, Switzerland. "Working out how humans behave is more interesting than whether it is adaptive or not."

Either way, there appears to be something deep within us that drives us to help others – even strangers. And if any of this suggests ways that fund-raisers might appeal to our altruistic tendencies when faced with the next humanitarian crisis, surely that is all to the good.

Why are we so generous?

1. Just for Kicks

In 2002, a team of researchers led by psychiatrist Gregory Berns from Emory University in Atlanta, Georgia, used brain imaging to find out what is going on inside our heads when we cooperate. They discovered that when players work together in the prisoner's dilemma game, the active parts of their brain include the orbitofrontal cortex and the striatum – areas associated with processing reward (*Neuron*, vol 35, p 395). And, last year, economist Ernst Fehr and psychologist Dominique de Quervain of the University of Zurich discovered that we get a similar mental buzz when we punish cheats, even when it means incurring a personal monetary cost (*Science*, vol 305, p 1254).

2. It's Good for the Image

Punishing others who don't toe the line can boost your reputation, as a recent study by anthropologists Rob Boyd and Karthik Panchanathan of the University of California at Los Angeles shows. Using computer simulations, they explored the benefits of a strategy of punishment that entails simply shunning others with a bad reputation and helping those with a good reputation. By doing this,

individuals can enhance their own standing, they found. What's more, by altering their behaviour according to people's reputations, these individuals minimise the cost of meting out punishment and gain the edge over indiscriminate cooperators who help anyone regardless of reputation (*Nature*, vol 432, p 499).

3. To Please Teachers (and Gods)

Despite our altruism, generosity may not be in our genes. If true altruism has evolved through competition between groups, as some researchers maintain, then it is more likely to be the product of cultural evolution. Genetic evolution works by selecting individuals with traits that are well adapted to their environment, but it has a far weaker grip on traits that benefit the group. So altruism is more likely to be learned. After all, every human culture invests considerable effort in instilling children with moral norms that help further cooperation. Often these are enshrined in powerful religious beliefs and reinforced by promises of salvation and threats of eternal damnation.

Topics: Evolution